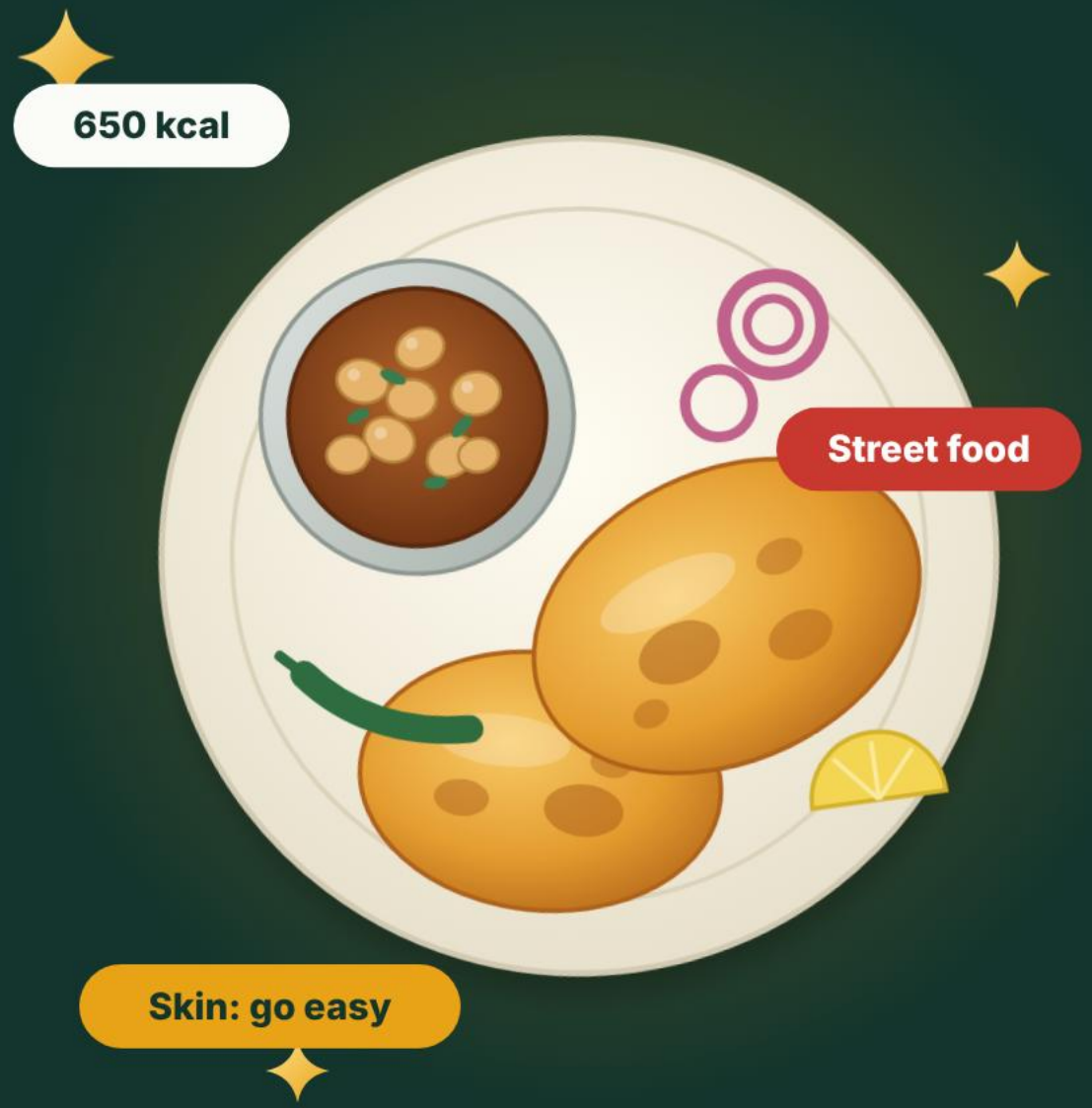
TCS DAY PROJECT

The Nutritionist

Your own personal nutritionist

Photograph a plate. See what it does to your body, your skin and your glow. Cook a better swap.



Built with my team at TCS Day

THE PROJECT

The Idea

Students in hostels and shared flats eat a lot of street food, and rarely see what it does to them.

The Nutritionist is a personal nutrition coach in your browser. It reads your plate, explains the effect on your body, and helps you cook a better swap.

8
features in one app

3
steps: see, understand, fix

1
photo to start



See it

Photograph a plate or a label. Get calories and nutrients.



Understand it

See the weekly effect on your gut, skin, liver and kidneys.



Fix it

Cook a home swap, build a street-free streak and earn rewards.

EXAMPLE: ONE PLATE OF CHOLE BHATURE

How It Works

01



Snap

Photograph a plate or a label.

Chole bhature

02



Understand

AI finds every item and adds up the nutrients.

650 kcal, 30 g fat

03



Swap

Cook a home swap for your diet and kitchen.

Dal khichdi, 340 kcal

04



Glow

Street-free days build a streak and a skin forecast.

Day 7: early glow

FEATURE 01

Plate Scan

1. Snap a plate, or the label on a packet.
2. An image model and Gemini name each item.
3. See calories, nutrients and a risk level.

EXPECTED OUTPUT · CHOLE BHATURE

650	30	900
kcal	g fat	mg sodium

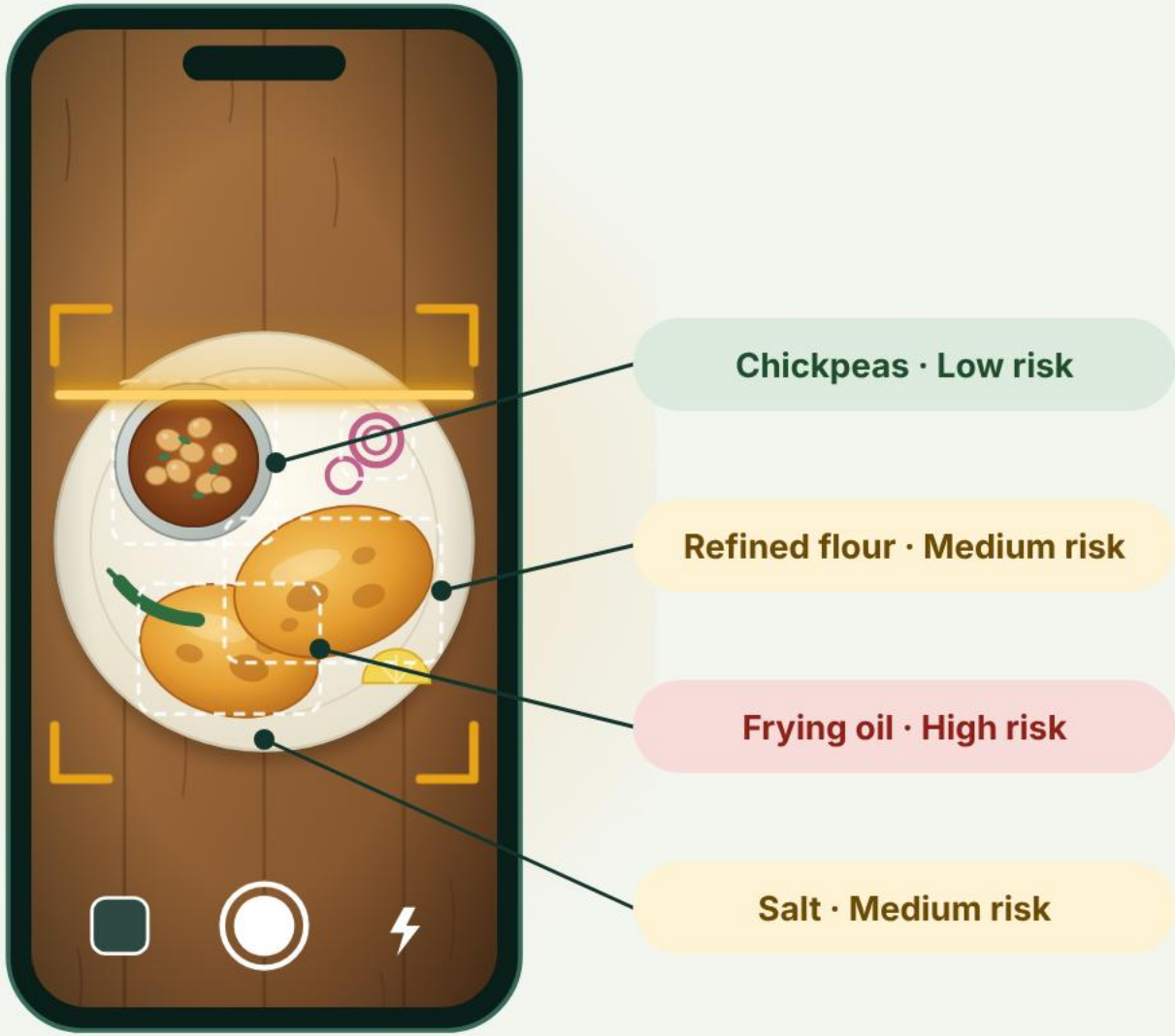


Illustration based on the app's own sample for chole bhature

FEATURE 02

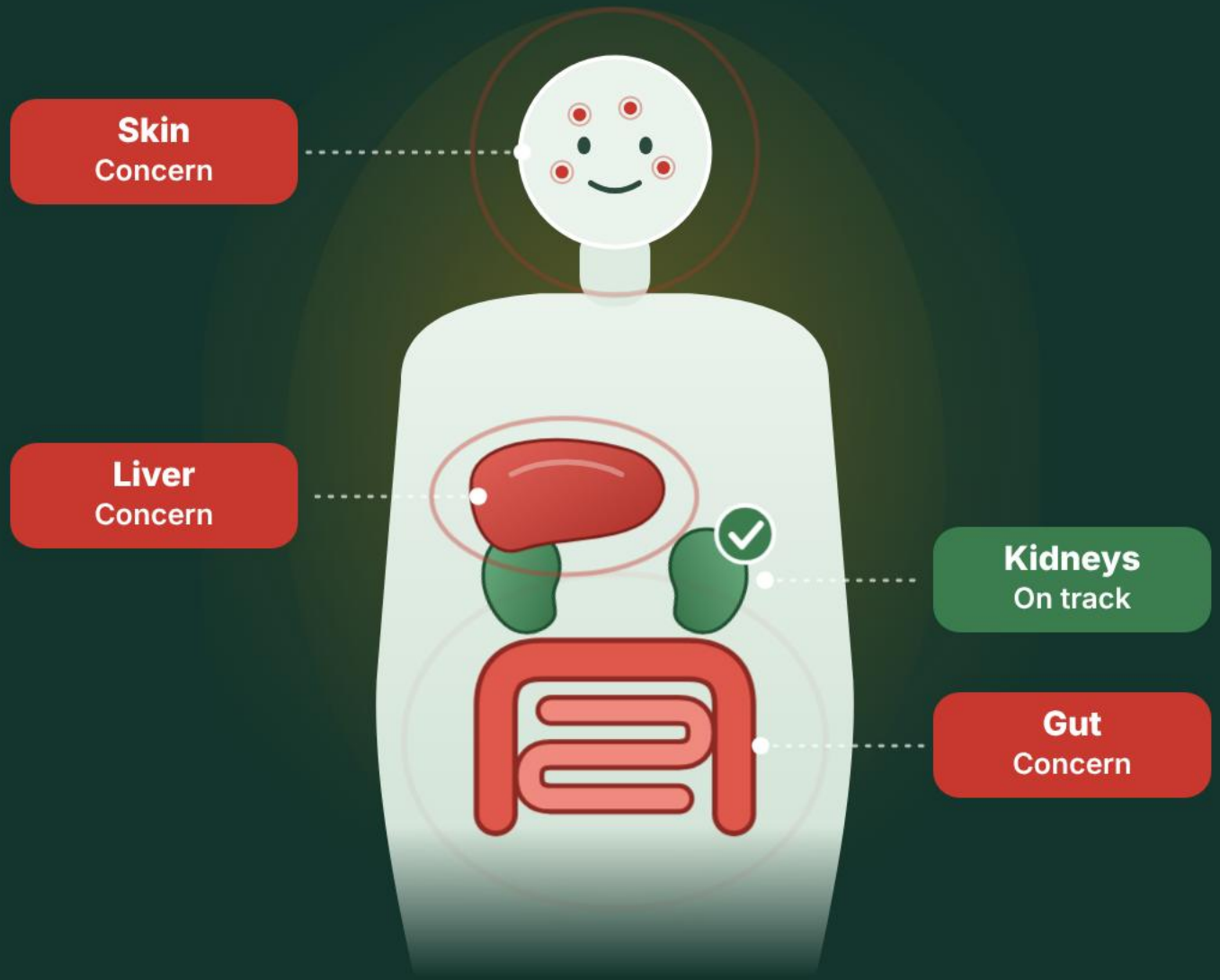
Body Impact

Your week, read by your body.

5	11 g	19 g
street or oily meals	fibre a day, aim 25 g	sugar a day

EXPECTED OUTPUT · A WEEK WITH FIVE STREET MEALS

Gut, Skin and Liver show Concern. Kidneys stay On track at 951 mg of sodium a day.



Sample week of logged meals. General guidance, not medical advice.

FEATURE 03

Food Chat

1. Open the chat on any scanned meal.
2. Ask in plain words, or tap a quick question.
3. Get a short answer in plain language.



Ask about this food

Chole bhature (1 plate)

Illustrative example

Is it bad for my liver?

Fried bhature add a lot of oil, which your liver has to process. It is fine now and then. Try it with salad or curd, and keep it occasional.

Is it bad for my liver?

Is this enough protein for me?

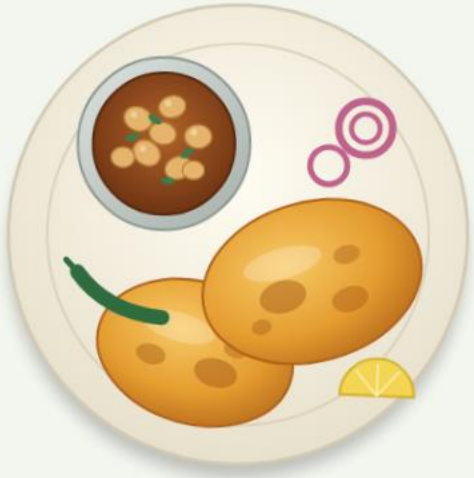
What can I eat instead?

Is it bad for my skin or kidneys?

Example answer for illustration. The app's replies vary.

FEATURE 04

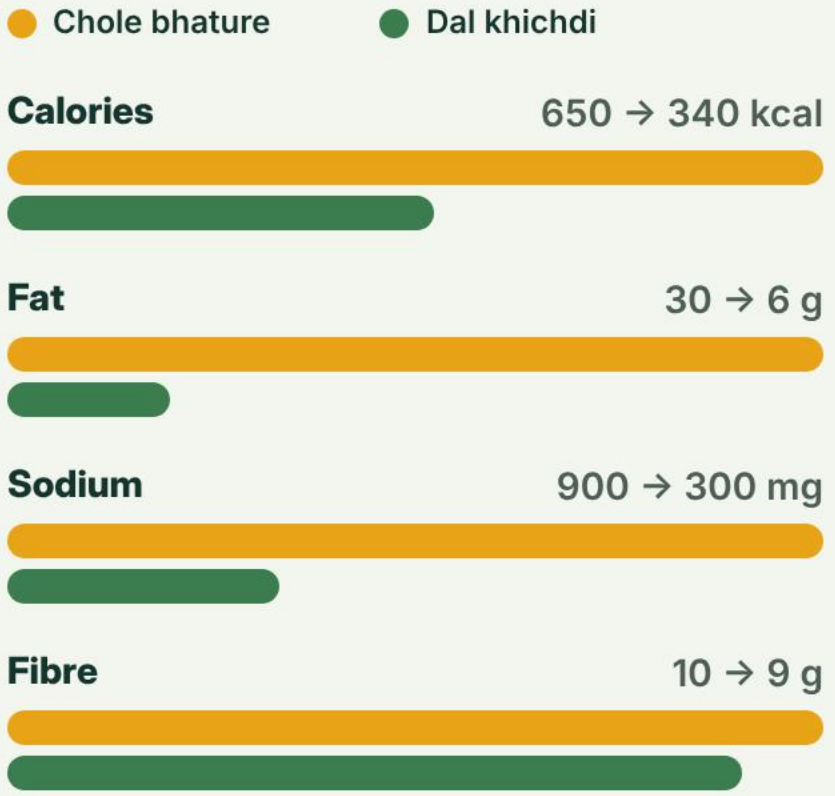
Cook Instead



Chole bhature
1 plate, street food



Dal khichdi
One-pot, 25 minutes



Half the calories, a fifth of the fat, a third of the salt, and about the same fibre.

Numbers from the app's own sample data for chole bhature and dal khichdi

FEATURE 05

Pantry Meals

Tell it what is in your kitchen.



Chana



Onion



Tomato



Curd

AI suggests

Three meals, chosen to suit your skin type.



Masala chana bowl

Skin-friendly

Chana, onion, tomato



Tawa chana tikki

Skin-friendly

Chana, onion



Chana raita bowl

Skin-friendly

Chana, curd, onion

Illustrative example. The AI's suggestions change every time.

NEW IN THIS VERSION

Skin and Glow

Your food, seen through your skin.

06 Skin type

07 Weekly skin score

08 Glow forecast



FEATURE 06 NEW

Skin Type

Pick your skin type once. Every meal gets a skin note, and the AI suggestions are chosen to suit it.

Example: chole bhature



Oily

Go easy

Higher in oil.



Dry

Go easy

Salty. Drink water with it.



Combination

Go easy

Higher in oil. May show in the T-zone.



Normal

Go easy

Fried. Fine now and then.



Sensitive

Go easy

Heavy or sugary. Can flare skin.

EXPECTED OUTPUT · OILY SKIN

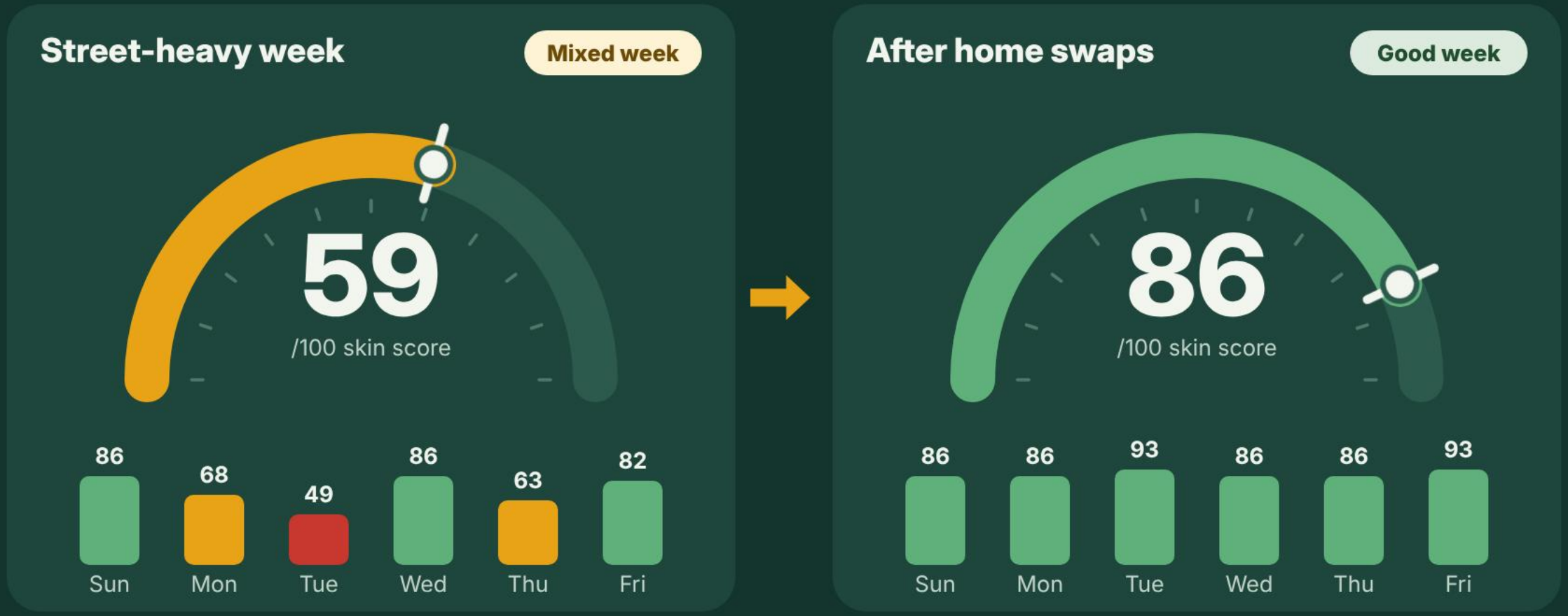
Eat more: dal, sprouts, chana, papaya, guava. Go easy on: samosa, pakora, bhature, fries.

Verdicts are the app's own, shortened. General guidance.

FEATURE 07 NEW

Weekly Skin Score

Sugar, oily meals, fibre, salt and glow foods, weighted for your skin type.



Two sample weeks, scored for oily skin. General guidance, not medical advice.

FEATURE 08 NEW

Glow Forecast

Your 7-day streak now comes with a skin forecast.

You are here



Day 1–2

Less puffy

Face may look less puffy.



Day 3

Steadier skin

Oil and inflammation may settle.



Day 5

Fresher mornings

Skin may look fresher on waking.



Day 7

Early glow

Brighter, calmer skin. Clearing takes longer.



Day 14

Clearer

Breakouts and dullness often start to ease.



Day 28

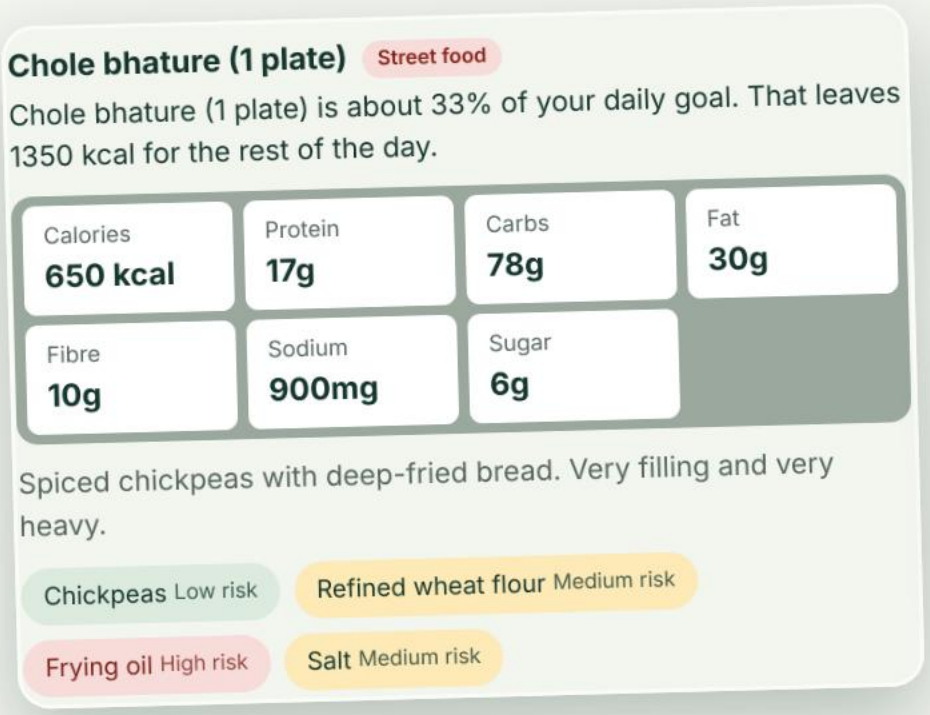
A full skin cycle

Fewer breakouts and a more even tone.

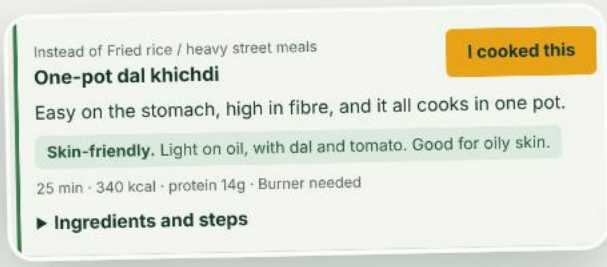
Typical expectations, not promises. See a dermatologist for persistent skin problems.

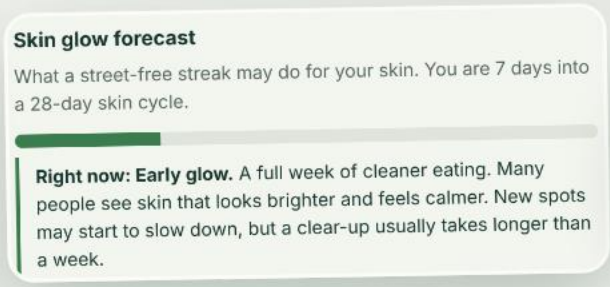
INSIDE THE APP

Real Screens









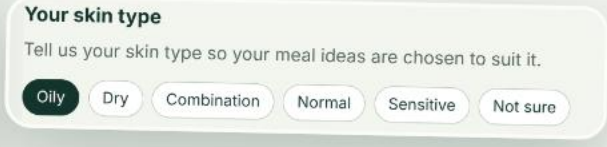


Plate scan · Body impact · Cook instead · Skin type · Glow forecast

Screens from the running app, using its built-in sample data.

TECH

Under the Hood



Photo

A plate or a packet label

Image model

ViT on Food-101 guesses the dish

Gemini vision

Names each item and estimates portions

Health engine

Plain Python adds up totals, scores and flags

Your app

Dashboard, history, swaps and streaks

THE RULE

The AI never has the last word on numbers.
Totals, targets and scores are calculated in code.

FastAPI · Gemini · PyTorch · Hugging Face · SQLite · PostgreSQL · Plain JavaScript

TCS DAY PROJECT

Built with my team at TCS Day.

Thank you for reading. What should The Nutritionist learn next?

ALSO IN THE APP

- Product label scanner
- History with CSV export
- Health profile and daily targets
- Points, levels and badges

NEXT UP

A skin note on every scanned meal
Skin type saved to your account
A live demo link



The Nutritionist

Your own personal nutritionist